

MiniMBA for Engineers : Business Health Analytics





Goal of a Business

1) Profit Maximization?

this goal ignores:

a) **TIMING** of Returns

(Time Value of Money)

b) **UNCERTAINTY** of Returns

(Risk)

Goal of a Business

2) Shareholder Wealth Maximization?

this is the same as:

- a) Maximizing Firm Value**
- b) Maximizing Stock Price**

The Corporation and Financial Markets

Corporation



The Corporation and Financial Markets

Corporation



Investors



The Corporation and Financial Markets

Corporation

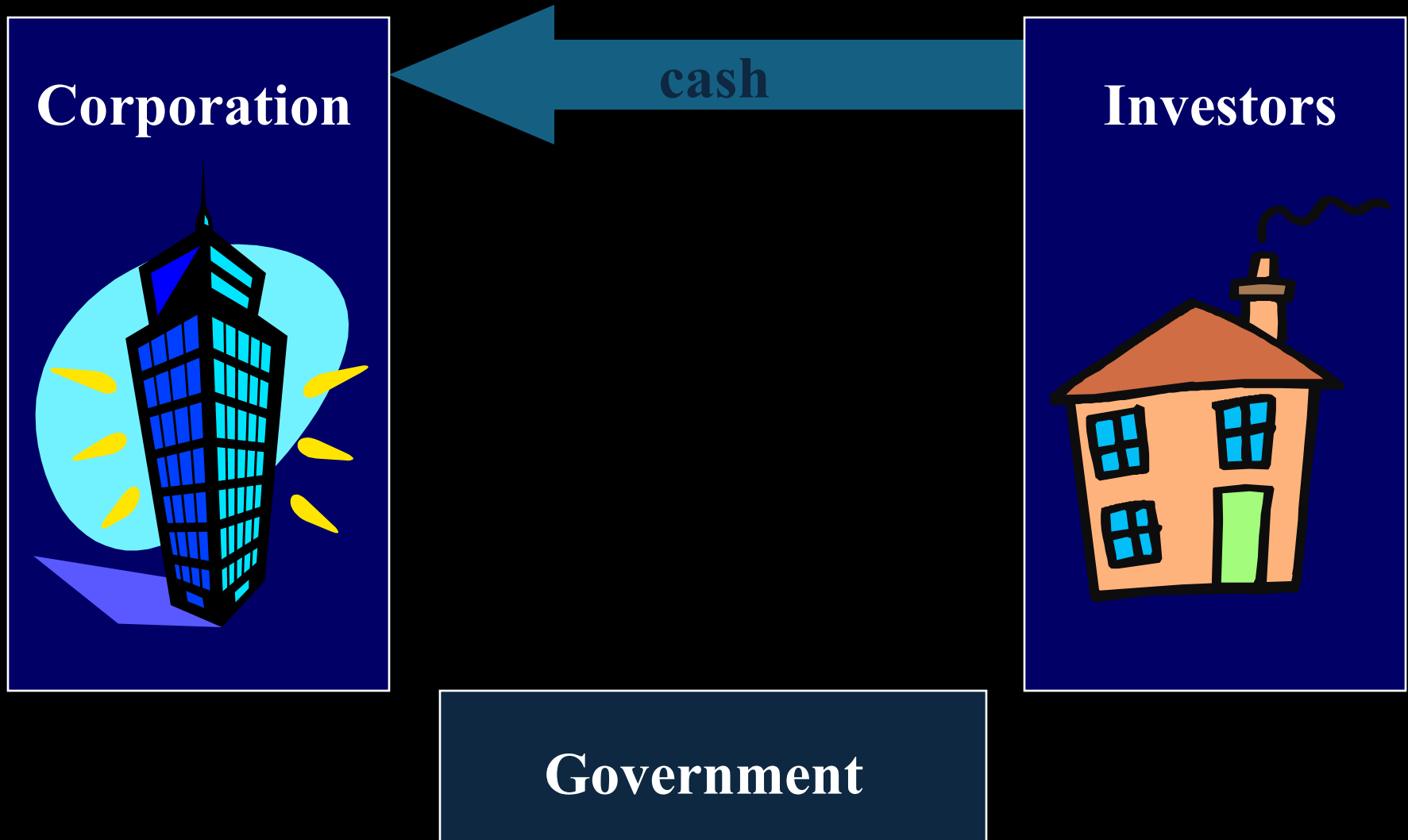


Investors

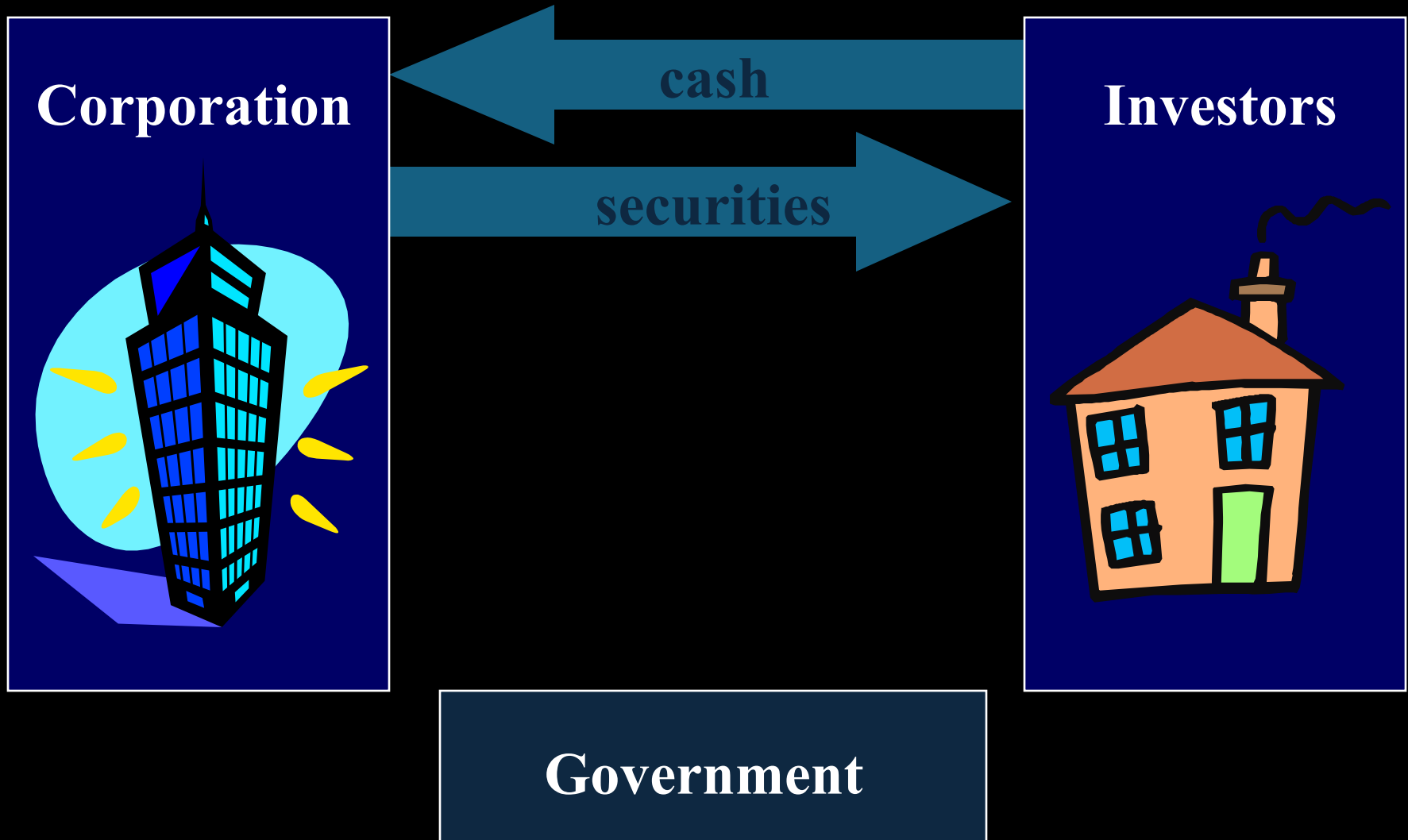


Government

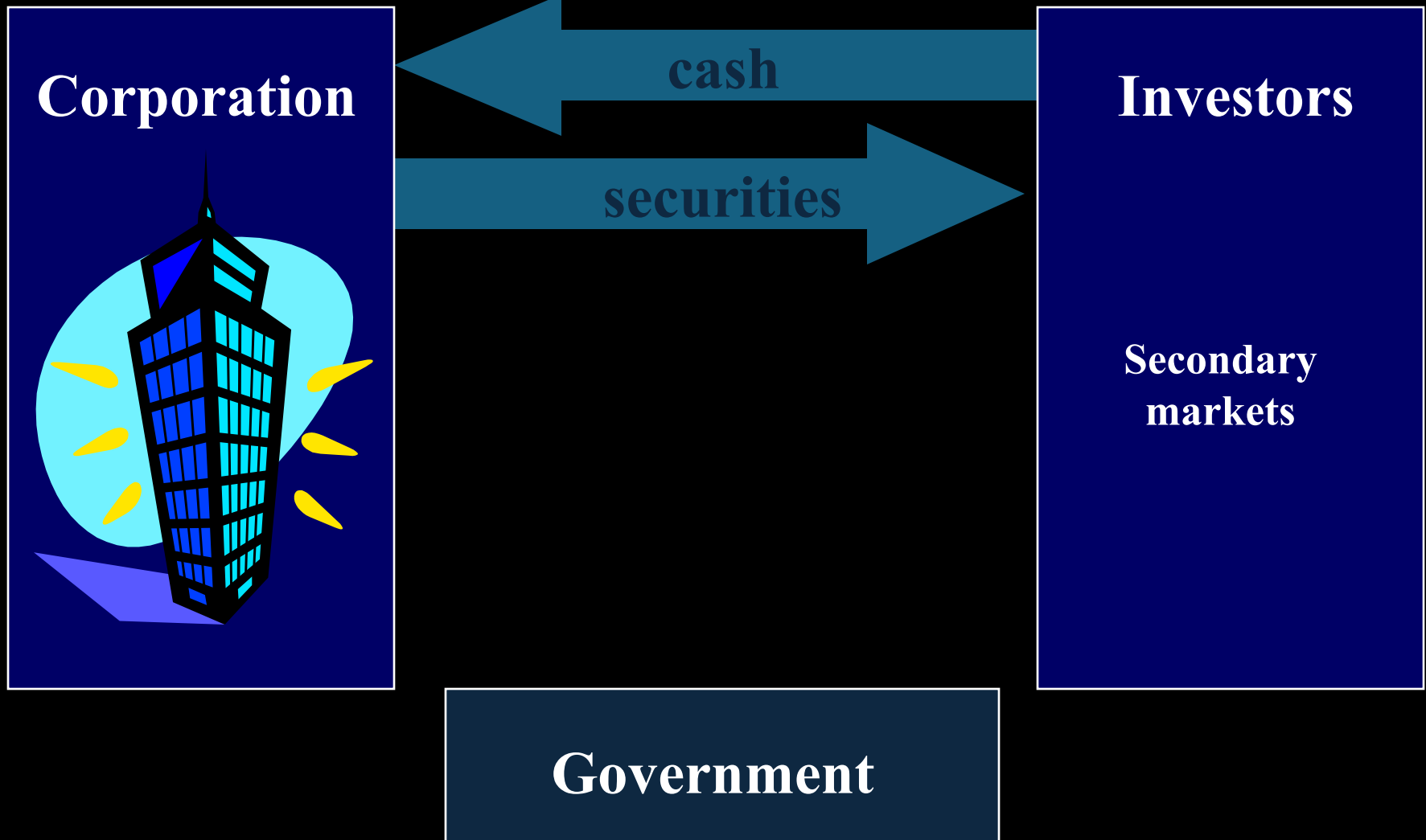
The Corporation and Financial Markets



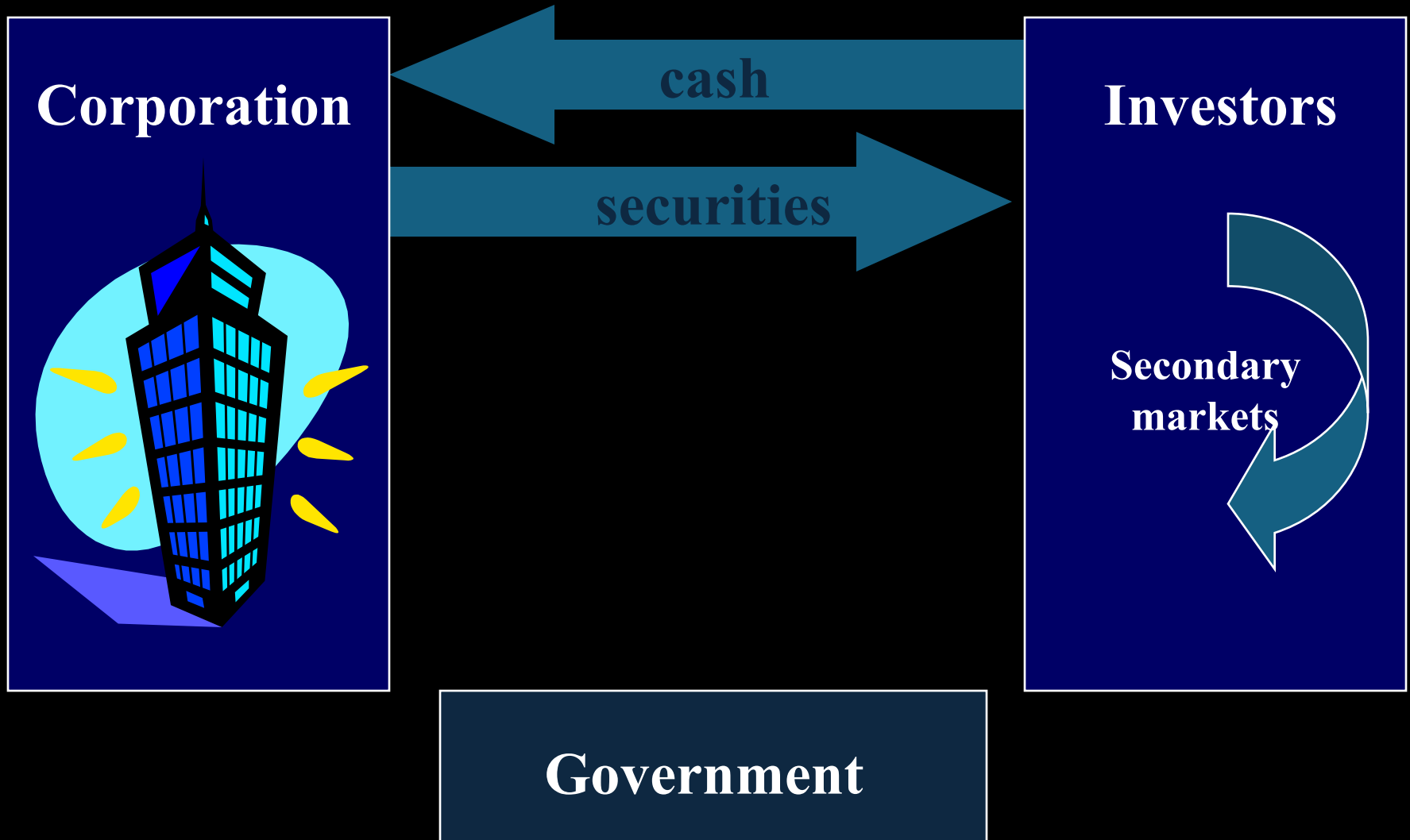
The Corporation and Financial Markets



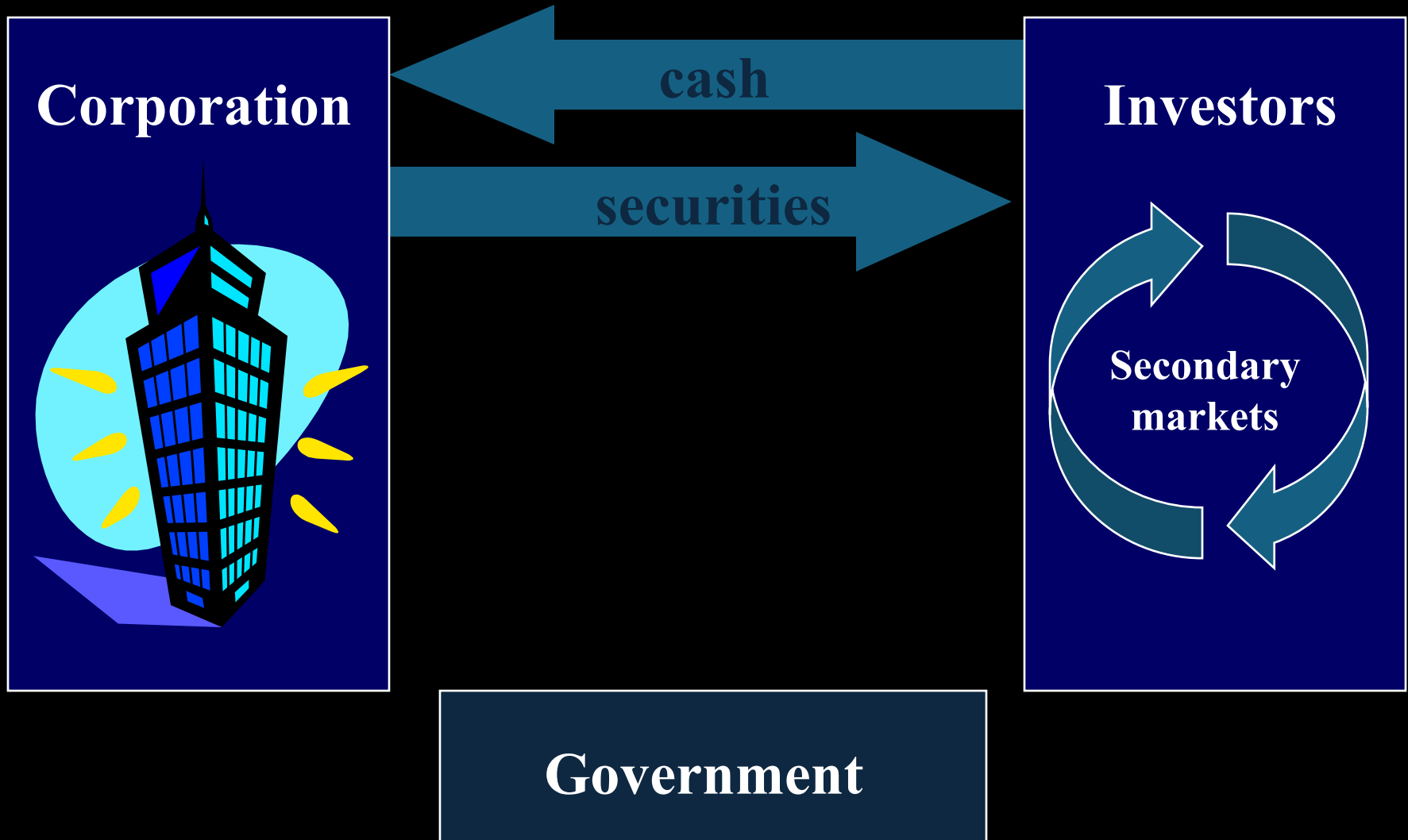
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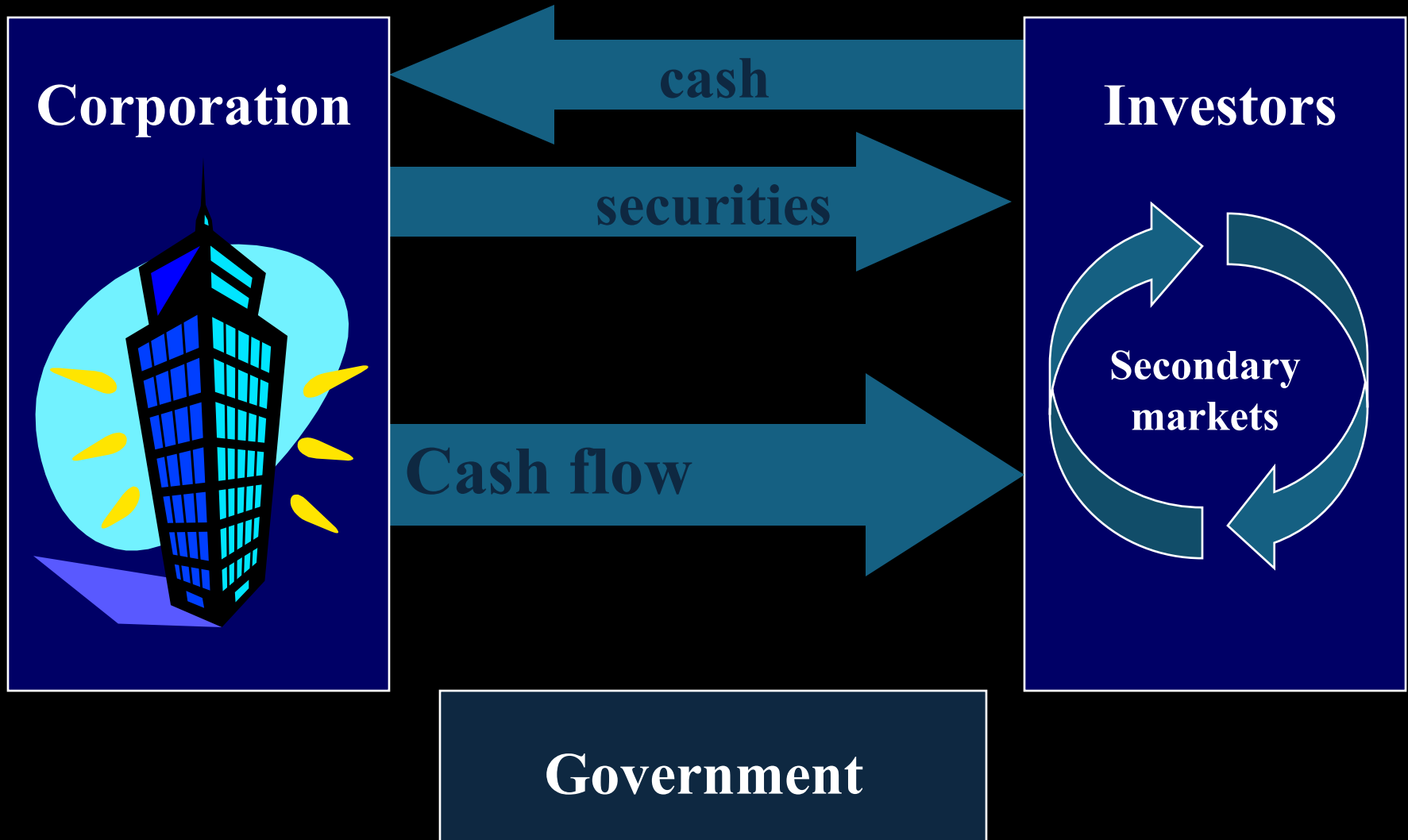
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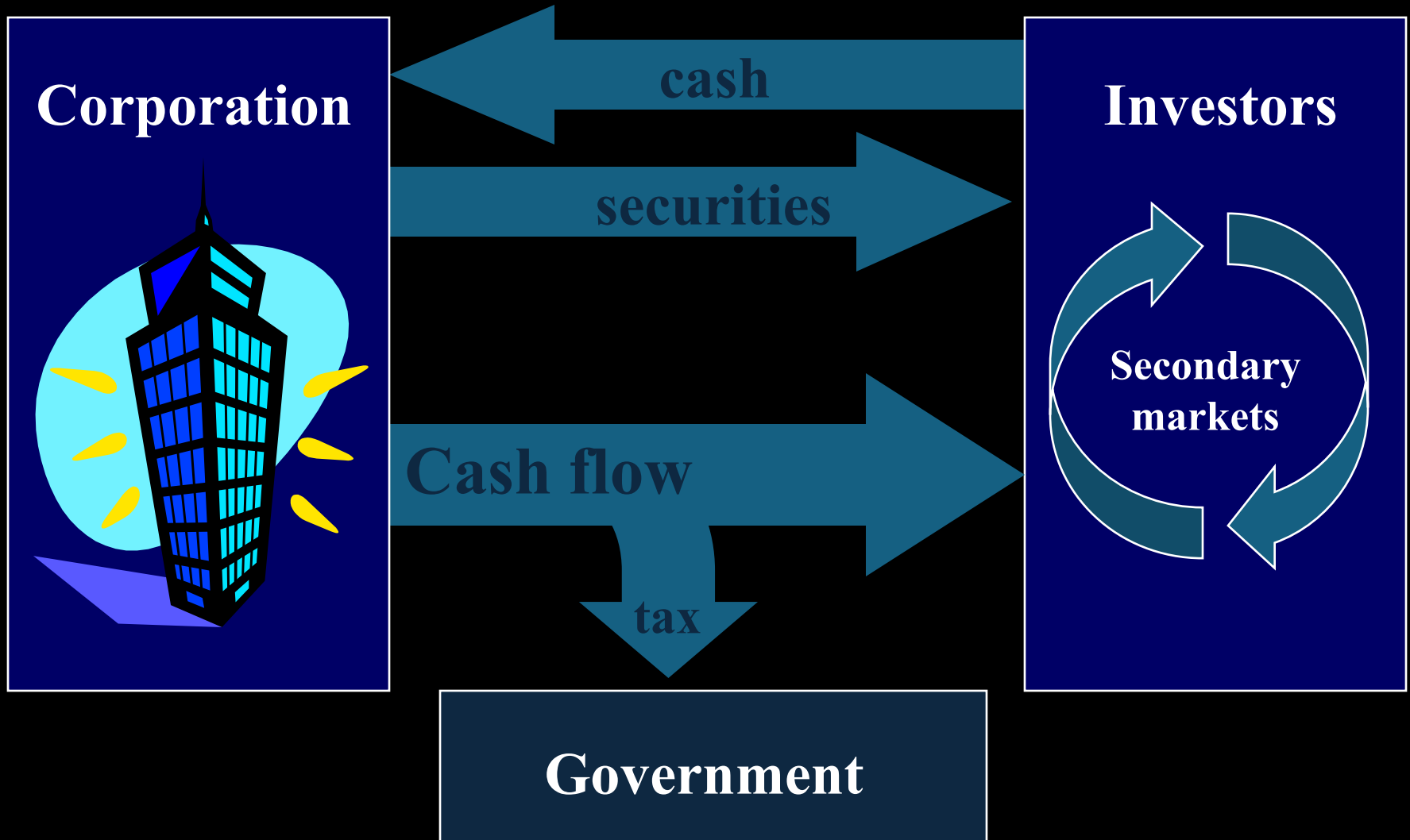
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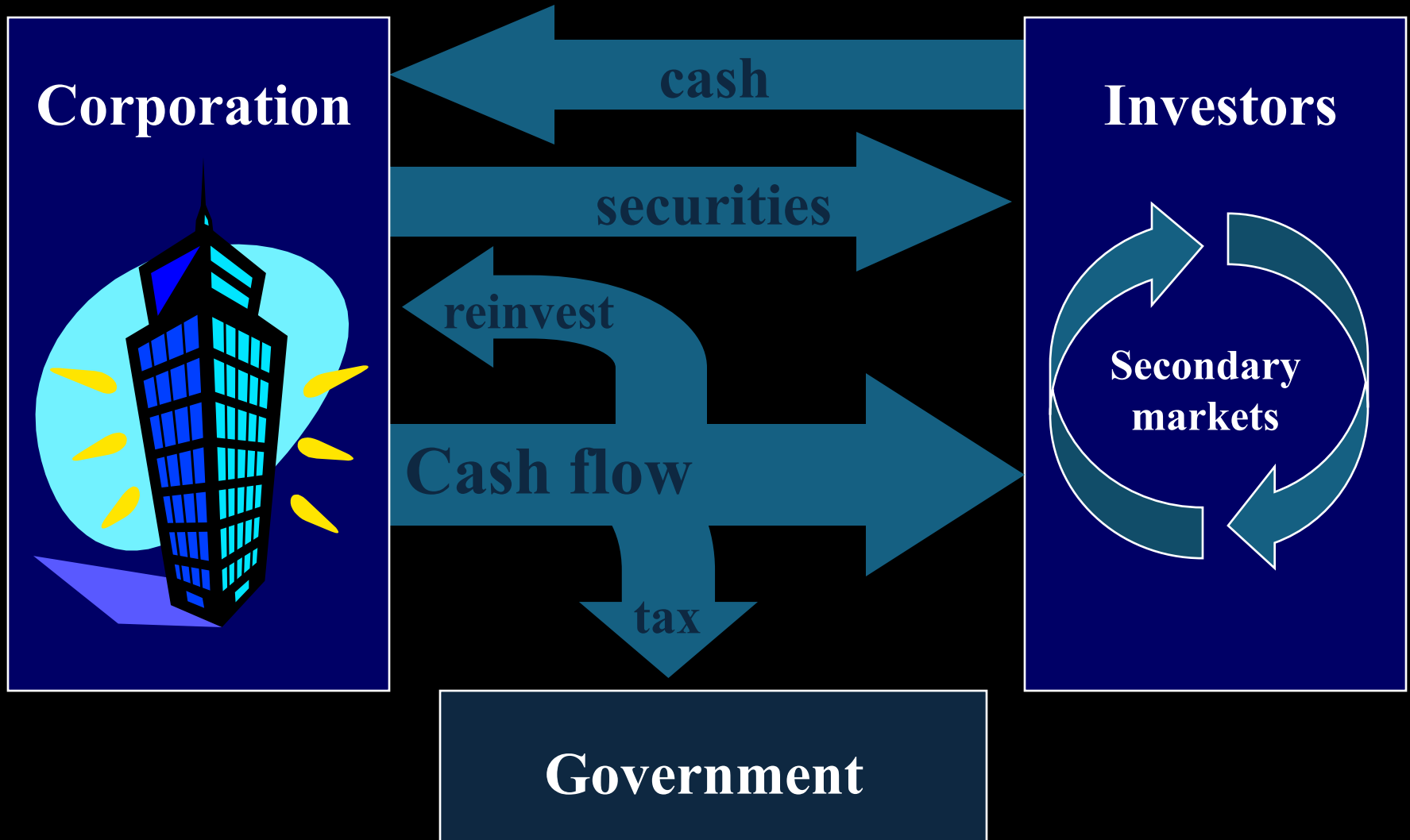
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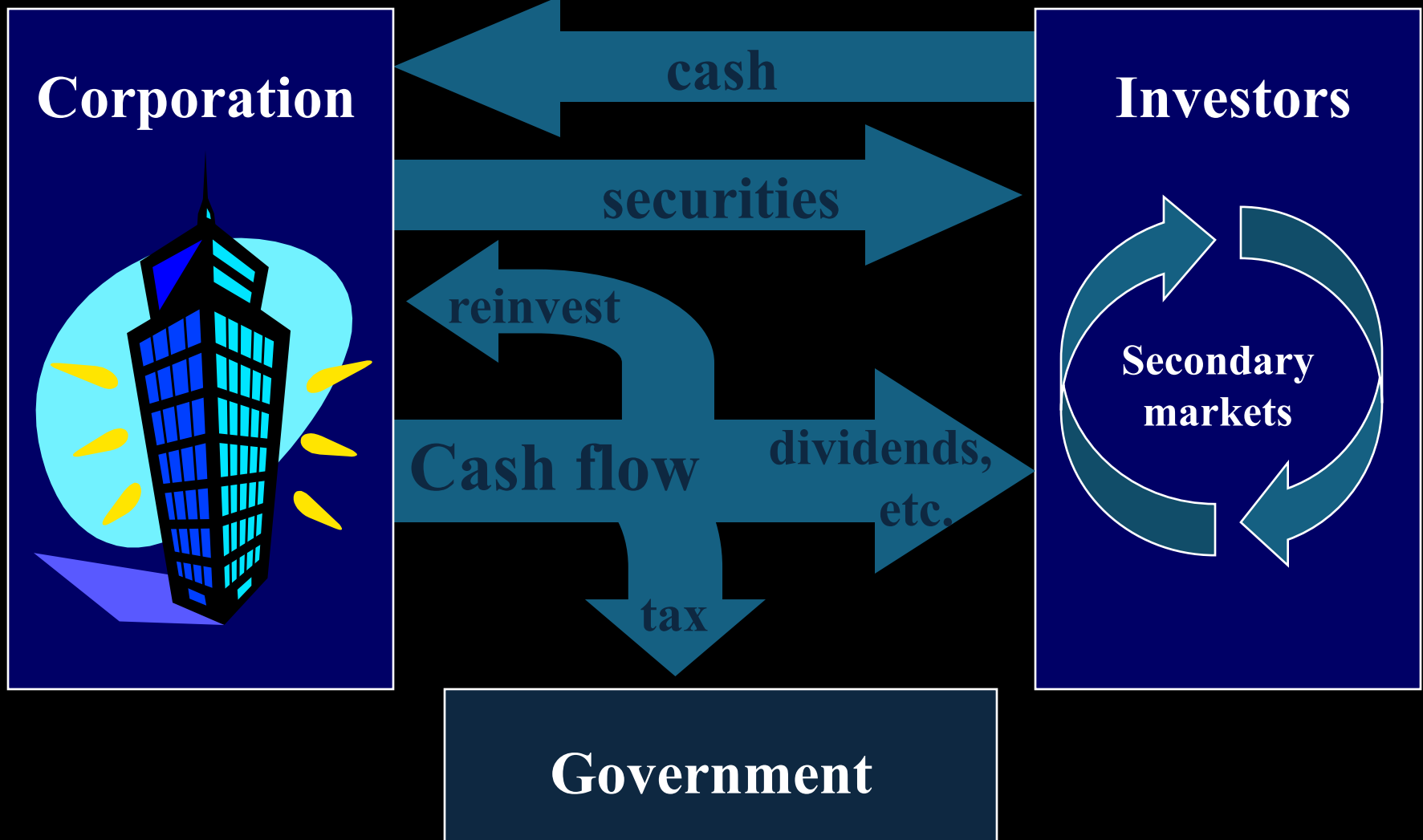
The Corporation and Financial Markets



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The Corporation and Financial Markets

- Primary Market

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- **Primary Market**
 - Market in which new issues of a security are sold to initial buyers.

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- **Primary Market**
 - Market in which new issues of a security are sold to initial buyers.
- **Secondary Market**

The Corporation and Financial Markets

- **Primary Market**

- Market in which new issues of a security are sold to initial buyers.

- **Secondary Market**

- Market in which previously issued securities are traded.

The Corporation and Financial Markets

- Initial Public Offering (IPO)

The Corporation and Financial Markets

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 - The first time the firm's stock is sold to the general public.

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- Seasoned New Issue

The Corporation and Financial Markets

- **Initial Public Offering (IPO)**
 - The first time the firm's stock is sold to the general public.
- **Seasoned New Issue**
 - A new stock offering by a firm that already has stock that is traded in the secondary market.

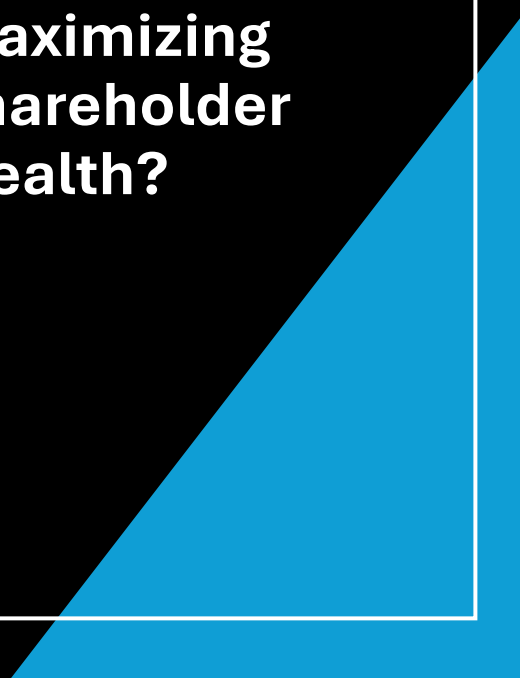
Financial Management Axioms

- 1) Risk - return trade-off.
- 2) Time value of money.
- 3) Cash - not profits - is king.
- 4) Incremental cash flows count.
- 5) The curse of competitive markets.
- 6) Efficient capital markets.
- 7) The agency problem.
- 8) Taxes bias business decisions.
- 9) All risk is not equal.
- 10) Ethical dilemmas are everywhere in finance.

Evaluating a business' Financial Performance

Financial Ratio Analysis

**Are our
decisions
maximizing
shareholder
wealth?**



Financial Ratios

- Tools that help us determine the financial health of a company.
 - We can compare a company's financial ratios with its ratios in previous years (trend analysis).
 - We can compare a company's financial ratios with those of its industry.
- 
- A large blue right-angled triangle is positioned in the bottom right corner of the slide, pointing towards the top left.

**We will want to answer questions
about the firm's**

- **Liquidity**
- **Efficient use of Assets**
- **Leverage (financing)**
- **Profitability**

Example: BigBear Corporation



BigBear's Balance Sheet (\$000)

Assets:

Cash	\$2,540
Marketable securities	1,800
Accounts receivable	18,320
Inventories	27,530
Total current assets	50,190
Plant and equipment	43,100
less accum deprec.	11,400
Net plant & equip.	<u>31,700</u>
Total assets	81,890

Liabilities & Equity:

Accounts payable	9,721
Notes payable	8,500
Accrued taxes payable	3,200
Other current liabilities	4,102
Total current liabilities	25,523
Long-term debt (bonds)	<u>22,000</u>
Total liabilities	47,523
Common stock (\$10 par)	13,000
Paid in capital	10,000
Retained earnings	11,367
Total stockholders' equity	<u>34,367</u>
Total liabilities & equity	81,890

BigBear's Income Statement

Sales (all credit)	\$112,760	
Cost of Goods Sold		<u>(85,300)</u>
Gross Profit		27,460
Operating Expenses:		
Selling	(6,540)	
General & Administrative	<u>(9,400)</u>	
Total Operating Expenses		<u>(15,940)</u>
Earnings before interest and taxes (EBIT)	11,520	
Interest charges:		
Interest on bank notes:	(850)	
Interest on bonds:	<u>(2,310)</u>	
Total Interest charges		<u>(3,160)</u>
Earnings before taxes (EBT)	8,360	
Taxes (assume 40%)		<u>(3,344)</u>
Net Income		<u>5,016</u>

Other Information

Dividends paid on common stock	\$2,800	
Earnings retained in the firm	2,216	
Shares outstanding (000)		1,300
Market price per share	20	
Book value per share	26.44	
Earnings per share		3.86
Dividends per share		2.15

Liquidity Ratios

**Do we have
enough liquid
assets to meet
approaching
obligations?**

Current
Ratio

current assets
current liabilities

What is BigBear's Current Ratio?

$$\frac{50,190}{25,523} = 1.97$$

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$$\frac{50,190}{25,523} = 1.97$$

If the average current ratio for the industry is 2.4, is this good or not?

Current ratio

- The higher the current ratio, the more capable the company is of paying its obligations. A ratio under 1 suggests that the company would be unable to pay off its obligations if they came due at that point.

What is the Company's Acid Test Ratio?

A large, solid blue triangle is positioned in the bottom right corner of the slide, pointing towards the top right.

What is the firm's Acid Test Ratio?

$$\frac{\text{current assets} - \text{inventories}}{\text{current liabilities}}$$

What is the firm's Acid Test Ratio?

$$\frac{50,190 - 27,530}{25,523} = .89$$

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$$\frac{50,190 - 27,530}{25,523} = .89$$

**Suppose the industry average is .92.
What does this tell us?**

Acid Test Ratio

- Companies with ratios of less than 1 cannot pay their current liabilities and should be looked at with extreme caution. Furthermore, if the acid-test ratio is much lower than the working capital ratio, it means current assets are highly dependent on inventory. Retail stores are examples of this type of business.

**What is the
firm's
Average
Collection
Period?**



What is the firm's Average Collection Period?

**accounts receivable
daily credit sales**

What is the firm's Average Collection Period?

$$\frac{18,320}{112,760/365} = 59.3 \text{ days}$$

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$$\frac{18,320}{112,760/365} = 59.3 \text{ days}$$

**If the industry average is 47 days,
what does this tell us?**

Operating Efficiency Ratios

- Measure how efficiently the firm's assets generate operating profits.

What is the firm's Operating Income
Return on Investment (OIROI)?

$$\frac{\text{operating income}}{\text{total assets}}$$

**What is the firm's Operating Income
Return on Investment (OIROI)?**

$$\frac{11,520}{81,890} = 14.07\%$$

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- Slightly below the industry average of 15%.

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$$\frac{11,520}{81,890} = 14.07\%$$

- Slightly below the industry average of 15%.
- The OIROI reflects product pricing and the firm's ability to keep costs down.

**What is their
Operating Profit
Margin?**

A blue triangle graphic is located in the bottom right corner of the slide, pointing towards the top left.

What is their Operating Profit Margin?

$$\frac{\text{operating income}}{\text{sales}}$$

What is their Operating Profit Margin?

$$\frac{11,520}{112,760} = 10.22\%$$

What is their Operating Profit Margin?

$$\frac{11,520}{112,760} = 10.22\%$$

- **This is below the industry average of 12%.**

**What is
their Total
Asset
Turnover?**



What is their Total Asset Turnover?

$$\frac{\text{sales}}{\text{total assets}}$$

What is their Total Asset Turnover?

$$\frac{112,760}{81,890} = 1.38 \text{ times}$$

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$$\frac{112,760}{81,890} = 1.38 \text{ times}$$

**The industry average is 1.82 times.
The firm needs to figure out how to
squeeze more sales dollars out of its
assets.**

**What is the
firm's
Accounts
Receivable
Turnover?**



**What is the firm's Accounts Receivable
Turnover?**

**credit sales
accounts receivable**

What is the firm's Accounts Receivable Turnover?

$$\frac{112,760}{18,320} = 6.16 \text{ times}$$

What is the firm's Accounts Receivable Turnover?

$$\frac{112,760}{18,320} = 6.16 \text{ times}$$

BigBear turns their A/R over 6.16 times per year. The industry average is 8.2 times. Is this efficient?

**What is the
firm's Inventory
Turnover?**

A blue right-angled triangle is positioned in the bottom right corner of the slide, pointing towards the top left.

What is the firm's Inventory Turnover?

$$\frac{\text{cost of goods sold}}{\text{inventory}}$$

What is the firm's Inventory Turnover?

$$\frac{85,300}{27,530} = 3.10 \text{ times}$$

What is the firm's Inventory Turnover?

$$\frac{85,300}{27,530} = 3.10 \text{ times}$$

**BigBear turns their inventory over
3.1 times per year.**

The industry average is 3.9 times.

Is this efficient?

Low inventory turnover:

The firm may have too much inventory, which is expensive because:

- **Inventory takes up costly warehouse space.**
- **Some items may become spoiled or obsolete.**